

Unit 1: Android Operating System and Development Environment

Focus: Architecture, SDK, OHA, and Project Structure

- State the purpose of the Open Handset Alliance (OHA).
 - Mention the purpose of the Android SDK.
 - Define "Android Virtual Device" (AVD).
 - What is the role of the Dalvik Virtual Machine (DVM) in Android?
 - What is the significance of the AndroidManifest.xml file?
 - Explain advantages and limitations of Android operating system.
 - List any four features of Android.
 - Explain the role of the Android Development Tools (ADT).
 - Briefly describe any three Android versions and one key feature of each.
 - Explain the directory structure of an Android project.
 - Briefly explain Android Architecture with a Diagram. * Discuss the evolution of the Android operating system from its inception to the current version.
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Unit 2: Android Components and Resource Handling

Focus: Activities, Services, Intents, and Resources

- Define an Activity in Android.
 - Localization supports multiple _____.
 - What is an Intent in Android?
 - What is localization in android? Explain importance of it.
 - Describe a Service in Android and state its use.
 - Scenario: A Login button is displayed on the screen but does not respond. How can you make it clickable using Java code?
 - Distinguish between "Styles" and "Themes" in Android.
 - Discuss the following Android components: Activity, Service, Broadcast Receiver, and Content Provider.
 - Explain the Activity Lifecycle in detail with the help of a state diagram.
 - Describe the process of "Localization" and how to prepare an application for multiple languages.
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Unit 3: Android User Interface Elements and Layouts

Focus: UI/UX, Material Design, and View Groups

- What is a View?
- Material Design improves the _____ of apps.
- Define the term "UX" (User Experience).

- Name any two types of Menus available in Android.
 - Explain the difference between LinearLayout and RelativeLayout.
 - What are the common properties and methods of a Button widget?
 - Describe the use of Dialog boxes in an Android application.
 - Illustrate Constraint Layout and state its advantages.
 - Explain the concept of "Material Design" and how it influences modern Android UI. * Discuss "Dynamic Implementation of Layout" and when it is preferred over XML-based layouts.
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Unit 4: Working with Views and Fragment

Focus: Lists, Grids, Adapters, and Fragments

Define a Fragment in Android.

- RecyclerView is used to display _____ items.
- What is the purpose of a WebView?
- When would you use a ScrollView?
- What is RecyclerView?
- Explain GridView and mention its use in Android applications.
- What is the role of a CardView in designing modern lists?
- Differentiate between ListView and RecyclerView with suitable points.
- Summarize the Fragment Life Cycle with the help of a diagram.
- * Explain how to implement a Fragment within an Activity and discuss its advantages in multi-pane layouts.